ROHIT KUMAR DUBBAKA

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About Me

Electronics & Communication graduate (2026) with exposure and contributions across the embedded-IoT development lifecycle - schematic design, PCB layout, fabrication, firmware stack, cloud integration and field deployment. Looking to contribute,perform and grow in the embedded systems landscape where system level ownership matters.

Experience

DeltaIOT Pvt Ltd — Feb 2026 - Present
Embedded Systems Intern — *Hyderabad*

- Delivered a compact 4-Layer LTE/GPS module board capable of handling 3A current bursts, contributing to the company's main-stream product.
- Delivered the full PCB design for a IC-based hot-swappable power management breakout board for 2.5A current bursts - from schematic to production
- Conducted hardware R&D by reverse-engineering an industry-grade IoT device pin-by-pin; enabling internal reference and component sourcing decisions.
- Participated in end-to-end IoT system deployments including field installation and a subsequent firmware OTA update, gaining exposure to real-world device lifecycle management.

Technical Skills

Hardware: ESP32, STM32, Analog & Digital Circuits
Tools: Altium Designer, KiCAD, STM32CubeIDE, ESP-IDF, Xilinx Vivado
Domains: PCB Design, Embedded Systems, Analog/Digital Circuits, Basic VLSI
Languages: C, Python, Verilog/VHDL

Projects

> Flagship Builds - *Engineered from scratch*

Hands-Free sEMG Based Speech-Controlled Wheelchair | *STM32, Python, LTspice, KiCAD* — *Jun 2025 - Mar 2026*

- Designed a custom analog front-end (AFE) PCB for sEMG acquisition from laryngeal muscles, validated for gain and CMRR on bench.
- Implemented real-time closed-loop firmware on STM32 integrating ADC, timers, and UART; trained a Random Forest classifier for muscle-command recognition.
- Delivered a full end-to-end system: wearable AFE → signal acquisition → ML inference → wheelchair actuation with safety overrides.

Priority Arbiter for Multi-Channel Bus | *VHDL, Basys-3 FPGA* — *Dec 2024 - Mar 2025*

- Designed and validated a deterministic multi-channel bus priority arbiter from RTL to FPGA using encoder and sequencer-based arbitration logic.
- Verified timing correctness through RTL simulation and confirmed deterministic behavior under concurrent access on hardware.

> Personal Exploratory Builds - *curiosity-driven.*

Aeromesh - CraftifAI Buildathon | *FirmGen, Esp-IDF, ESP32* — *Aug 2026*

- Rapid-prototyped firmware with FirmGen on a bare ESP32 + MPU6050, processing and detecting turbulence in-situ to avoid streaming chunks of raw sensor data.
- Trained an edge AI model within minutes during the buildathon to classify turbulence from live motion data.
- Built a diagnostic module for flight systems, designed to mesh with nearby devices and share real-time wind/air-situation data.

Automatic Plant Watering System | *Arduino Uno, C* — *Mar 2025*

- Used a solar panel as an ambient light sensor (ADC) with a custom scheduling algorithm to replace an RTC and trigger two daily watering cycles
- Actuated a relay-driven solenoid valve; serial logging enabled real-time monitoring when connected.

Analog Front-End for sEMG Signal Acquisition | *LTspice, KiCAD* — *Feb 2025 - Apr 2025*

- Designed a wearable-size PCB (60×25 mm) for sEMG signal acquisition; bench-validated gain and CMRR performance with surface electrodes.

Please refer my Github for all of my works

- https://github.com/rohit29d

Certifications

Links

PCB Design with KiCad - Peter Dalmaris (Udemy) (Aug 2026)

Machine Learning Techniques in MATLAB - MathWorks (Jun 2025)

SystemVerilog Fundamentals - Kumar Khandagle (Udemy) (Dec 2024)

Education

Amrita Vishwa Vidyapeetham, Coimbatore 2026
B.Tech in Electronics and Communication Engineering *CGPA: 7.6/10*

VINJEE Junior College, Hyderabad 2022
Class XII - Telangana Board of Intermediate Education

The Hyderabad Public School, Begumpet 2020
Class X - ICSE